Hitesh Arora

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Education

Thapar Institute of Engineering and Technology Patiala, Punjab, India
Bachelor of Engineering in Computer Engineering *Aug 2024 – June 2028*

- **Relevant Coursework:** Probability & Statistics, Data Structures & Algorithms, Machine Learning, Database Management Systems, Object-Oriented Programming (C++), Linear Algebra, Discrete Mathematics

Technical Skills

Languages: C++, Python, SQL, R, MATLAB, C
ML & Data: NumPy, Pandas, Scikit-learn, SciPy, TensorFlow, PyTorch, Matplotlib, Seaborn, OpenCV
Quant Finance: Algorithmic trading, backtesting, time-series forecasting, statistical arbitrage, options pricing, Black-Scholes model, Monte Carlo simulation, risk modeling, Sharpe ratio, portfolio optimization, alpha generation
Mathematics: Probability & Statistics, Stochastic Processes, Linear Algebra, Calculus, Combinatorics, Numerical Methods
Tools: Git, Jupyter Notebook, Linux, LeetCode (C++)

Projects

ML-Based Surface Crack Detection | *Python, OpenCV, Scikit-learn, NumPy* | Tata Steel 2025

- Designed a binary classification pipeline to detect surface cracks in steel sheets, achieving **91% test accuracy** on an imbalanced industrial dataset with a 1:4 class ratio.
- Reduced false-negative rate by **18%** over baseline by implementing SMOTE resampling, morphological image transforms, and texture-based feature engineering via OpenCV.
- Delivered model validation results and performance metrics to **Tata Steel engineering stakeholders**, accelerating adoption of automated defect detection across **3 production lines**.

Multivariate Regression Modeling – Housing Price Prediction | *Python, Pandas, Scikit-learn*Jan 2026 – Present

- Engineered **15+ predictive features** through correlation analysis, variance inflation factor (VIF) testing, and recursive feature elimination (RFE) to maximize signal quality.
- Boosted model accuracy by **22.5%** (R^2: 0.71 $\rightarrow$ 0.87) by benchmarking Linear Regression against Random Forest and tuning **8 hyperparameters** via GridSearchCV.
- Confirmed out-of-sample generalization across **5-fold cross-validation** with under **2% train-test $\mathbf{R^2}$ variance** – methodology transferable to walk-forward backtesting frameworks.

Time-Series Forecasting – Air Quality Index Prediction | *Python, Pandas, Scikit-learn* Jan 2026 – Feb 2026

- Built and optimized time-series regression models forecasting AQI from **3 years** of historical pollution and meteorological data, reducing RMSE by **22%** through lag-feature engineering.
- Uncovered autocorrelation structure in model residuals via ACF/PACF analysis, driving a further **15% reduction** in prediction error – technique applicable to trading alpha signal decay modeling.
- Generated **10+ diagnostic visualizations** of prediction intervals and error distributions across models, communicating uncertainty in formats analogous to financial risk reporting dashboards.

Certifications and Achievements

IBM Machine Learning Professional Certificate | IBM / Coursera 2026

- Completed a **12-course** professional program spanning supervised learning, ensemble methods, unsupervised learning, and end-to-end ML model deployment – reinforcing skills applied across all projects above.

Competitive Programming – LeetCode (C++) Ongoing

- Solved **100+** algorithmic problems in dynamic programming, greedy algorithms, and graph theory; skills transferable to low-latency system optimization in high-frequency trading environments.

Quantitative Finance – Independent Research Ongoing

- Investigated derivatives pricing frameworks (Black-Scholes, options Greeks, risk-neutral valuation) across **5+ foundational texts**, building rigorous theoretical grounding in financial mathematics.
- Modeled order book dynamics and execution algorithms (TWAP/VWAP) from **3+ market microstructure papers**, translating academic concepts into practical algorithmic trading intuition.